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UK Visas & Immigration
Global Talent Visa — Tech Nation (Digital Technology)
United Kingdom

To Whom It May Concern,

I write to provide my personal endorsement of Odeyemi Olusegun Israel in support of his application for the UK Global Talent Visa under the Digital Technology route. While my colleague Professor [CS Professor Name] is better placed to comment on the theoretical merits of Olusegun's research papers, my endorsement focuses on what I consider equally important: his **engineering capability, systems thinking, and the real-world impact of the products he has built**.

By way of background, I am a Professor of Electrical and Electronic Engineering at the University of Ibadan and have previously served as Director of Management Information Systems (MIS) at the same institution. In the MIS role, I was responsible for overseeing the university's entire digital infrastructure—enterprise systems, data management, and operational technology. My research spans power systems, computational intelligence, and applied machine learning in engineering contexts. This combination of institutional systems management and technical research gives me a particular perspective on what it takes to build complex, production-grade systems that actually work.

AMTTP: a product, not a prototype.

Olusegun has designed and built a system called the Anti-Money Laundering Transaction Trust Protocol (AMTTP). I want to be clear about what this is, because in my experience, most solo developers who claim to have built “platforms” have really built proof-of-concept demos. AMTTP is not that.

The system comprises four architectural layers that must function together in real time:

1. A **client SDK layer** (TypeScript and Python, both open-sourced on GitHub) that financial institutions integrate into their transaction pipelines. The TypeScript SDK alone exposes 19+ service endpoints—covering risk scoring, KYC, sanctions, dispute management, reputation, MEV protection, bulk operations (up

to 10,000 transactions per batch), webhooks, and geographic risk—with EIP-712 typed signing and HMAC-SHA256 webhook authentication.

2. A **compliance orchestration engine** comprising 7 parallel microservices (ML risk scoring, graph-based relationship analysis, policy enforcement, sanctions screening, continuous monitoring, geographic risk, and an explainability module) coordinated through a central API gateway with NGINX TLS termination, HS256 JWT validation, per-key rate limiting, and ECDSA oracle-signed responses with 90-day key rotation.
3. An **offline ML training pipeline** using a Composite Teacher architecture: XGBoost (weighted at 0.4), seven FATF-aligned AML pattern detectors (0.3), and graph structural properties (0.3) to generate pseudo-labels across 2.64 million Ethereum mainnet transactions sourced from Google BigQuery. A stacked ensemble of XGBoost (500 estimators) and LightGBM (500 estimators) with a logistic regression meta-learner achieves production ROC-AUC of **0.9879**, PR-AUC of **0.9715**, and F1 of **0.9012** with 21 features and zero training-serving skew. This is a notable engineering achievement in its own right: an earlier model version trained on 71 features but served with only 5 (66 features hardcoded to zero), producing an effective ROC-AUC of just 0.64. Diagnosing and correcting this lifted performance to 0.99 in a single iteration.
4. A **blockchain smart contract layer** with 18 Solidity contracts deployed on Ethereum Sepolia, handling on-chain risk attestation, risk-tiered transaction gating (Approve / Review / Escrow / Block), compliance audit trails, LayerZero cross-chain policy synchronisation, Kleros-integrated decentralised dispute resolution, Gnosis Safe and Biconomy account-abstraction modules, and Compliance NFTs.

The whole thing runs across 17 containerised microservices—each with health checks, circuit breaker patterns, and Prometheus/Grafana monitoring—orchestrated through Docker Compose in a production-grade configuration. I know, from my time managing the University of Ibadan’s IT systems, how difficult it is to keep even five or six services reliably communicating. Seventeen is a fundamentally harder problem, and Olusegun has solved it alone.

Zero-knowledge cryptography for regulatory compliance.

One component of AMTTP that I think deserves particular attention is zkNAF—Olusegun’s zero-knowledge Non-disclosing Anti-Fraud framework. It uses Groth16 ZK-SNARK proofs on the BN254 elliptic curve to allow institutions to prove four categories of compliance fact without revealing the underlying personal data: sanctions non-membership (Merkle non-membership proof without exposing the address or list contents), risk-range veri-

fication (ML score within acceptable bounds without revealing the exact value), KYC credential validity (completion without identity disclosure), and combined transaction compliance. On-chain, three dedicated verifier contracts are routed through a single `ZkNAFVerifierRouter`.

From a regulatory perspective, this is significant. Financial institutions currently face a genuine conflict between GDPR’s data minimisation requirements and AML regulations’ transparency demands. Olusegun’s solution does not merely acknowledge this tension—it resolves it at the cryptographic level. He has also implemented multi-oracle threshold signatures, server-enforced nonce + TTL (300 s) replay protection via Redis, and a UI Integrity Service that monitors DOM mutations with per-component hash verification—the kind of infrastructure-level security measures that separate production systems from academic exercises. The UI Integrity module was specifically designed to prevent address-swap attacks of the kind used in the February 2025 Bybit exploit, demonstrating that Olusegun stays current with live threat vectors.

Cross-platform consumer application.

Beyond the institutional infrastructure, Olusegun built a cross-platform Flutter consumer application targeting **six platforms** (iOS, Android, Web, Windows, macOS, Linux) from a single Dart codebase. The app includes a wallet dashboard with token and NFT management, a pre-transfer trust check with a traffic-light interstitial, transaction history, dispute filing, and MetaMask wallet connection. It communicates with the compliance backend through a hybrid Flutter / Next.js architecture using a bidirectional JavaScript Channel bridge. A separate Next.js 15 “War Room” dashboard serves institutional compliance analysts.

Building a production-quality consumer app on top of a 17-service backend is a substantial effort in itself. That Olusegun did it across six platforms, with real-time compliance integration, while simultaneously maintaining the backend and conducting research, speaks to an unusual level of sustained engineering output.

Why the architecture matters.

The core insight behind AMTTP addresses a problem that the entire blockchain compliance industry has, to date, failed to solve. In traditional banking, a suspicious transaction can be frozen mid-flight. On a blockchain, once a transaction reaches finality, it is irreversible. Every major compliance tool currently on the market—Chainalysis, Elliptic, TRM Labs—operates *after* settlement. They generate alerts; they do not prevent illicit transactions from completing. AMTTP intervenes *before* settlement, at the protocol level.

This is not simply a matter of calling an API earlier in the pipeline. It requires solving a

hard real-time systems problem: the ML inference, graph analysis, sanctions screening, and policy evaluation must all complete within the block confirmation window (approximately 12 seconds on Ethereum). Olusegun has engineered the system to meet these timing constraints while maintaining deterministic compliance guarantees. That is genuine systems engineering, not application development.

Interdisciplinary range and engineering maturity.

What distinguishes Olusegun, in my observation, is not just depth in any one area but the ability to work across domains at a high level simultaneously. AMTTP requires competence in:

- **Machine learning engineering:** training pipelines on 2.64M transactions, model deployment, feature engineering at scale, achieving ROC-AUC 0.9879.
- **Blockchain development:** 18 Solidity smart contracts, gas optimisation, on-chain/off-chain coordination, LayerZero cross-chain architecture, Kleros arbitration integration.
- **Distributed systems:** 17-service microservice orchestration, containerisation, API gateway design, circuit breaker patterns, observability.
- **Cryptographic engineering:** Groth16 zero-knowledge proofs, BN254 elliptic curve arithmetic, threshold signatures, secure key management, replay protection.
- **Cross-platform application development:** Flutter / Dart for 6 platforms, Next.js 15 for institutional dashboards, bidirectional JS Channel bridging.
- **Regulatory knowledge:** FATF Travel Rule, FCA requirements, OFAC / EU / UN sanctions compliance, GDPR data handling obligations.

In the university context, a project of this scope would typically require a principal investigator, two or three postdoctoral researchers, and a team of PhD students, with institutional support for infrastructure and compute. Olusegun has done it as a single engineer. I can confirm this because I have reviewed the GitHub repository (github.com/segetii/fintech) and the deployment configuration—there is no evidence of collaborators, outsourced components, or wrapper libraries pretending to be original work. It is all his.

The decision to open-source the client SDKs while keeping the orchestration core proprietary also reflects good product judgement. It lowers the integration barrier for potential adopters while protecting the core intellectual property. This is the kind of strategic decision that separates engineers who can build things from engineers who can build *products*.

Publications and intellectual property.

Olusegun has published the AMTTP architecture as a peer-reviewed preprint on TechRxiv (DOI: [10.36227/techrxiv.177220113.33816607/v1](https://doi.org/10.36227/techrxiv.177220113.33816607/v1)), his BSDT research on Zenodo (DOI: [10.5281/zenodo.19037687](https://doi.org/10.5281/zenodo.19037687)), the Universal Detection Principle on Zenodo (DOI: [10.5281/zenodo.19038783](https://doi.org/10.5281/zenodo.19038783)). On 4 March 2026, he filed a UK patent application with the Intellectual Property Office covering 25 claims across 13 invention groups—spanning the AMTTP compliance architecture, zkNAF, the ML detection framework, adaptive friction stabilisation, and the Navier-Stokes viscosity extension. Filing a patent with 25 claims as a solo applicant, covering technology that spans machine learning, cryptography, blockchain, systemic risk, and fluid dynamics, is itself an unusual achievement.

Cross-domain research impact.

Beyond AMTTP, Olusegun has produced a body of research work—Blind Spot Decomposition Theory, the Universal Detection Principle, and an Adaptive Friction Stability framework—that extends the engineering principles of AMTTP into original mathematical theory. I defer to my colleague’s letter for detailed comment on the theoretical merits, but I want to highlight one aspect from an engineering perspective: his third paper validates the adaptive friction framework across four domains—banking (6-quarter early warning before the 2008 GFC), algorithmic stablecoins (Terra/Luna collapse detected at hour 23 with simulation-reality correlation $r = 0.996$), energy infrastructure (Texas ERCOT blackout predicted 72 hours early with 71% capacity loss reduction), and 3D incompressible Navier-Stokes fluid dynamics.

As an electrical engineer with research in power systems, I can assess the ERCOT result directly: identifying gas supply chain disruption as the structural vulnerability, rather than frozen wind turbines (the public narrative), is the kind of insight that only emerges from a principled mathematical decomposition. The fact that the same mathematical framework that detects blockchain fraud also identifies power grid cascade origins is, frankly, remarkable.

Relevance to the UK digital economy.

The UK is actively positioning itself as a global leader in responsible fintech regulation. The FCA’s forthcoming framework for crypto-asset regulation will create substantial demand for compliance infrastructure that can operate at the protocol level. AMTTP is, as far as I am aware, the only system in existence that provides pre-settlement AML enforcement on blockchain transactions. The individual who built it would be a direct asset to the UK’s regulatory technology sector.

The cross-domain research—particularly the energy grid analysis and the Navier-Stokes work—also aligns with the UK’s strategic priorities in infrastructure resilience (Ofgem,

National Grid ESO) and applied mathematics (EPSRC, the Heilbronn Institute). Olusegun is not merely a fintech engineer; he is a systems thinker whose work has potential applications across multiple sectors of the UK economy.

My assessment.

In my years at the University of Ibadan, both as a professor and as the director responsible for institutional IT infrastructure, I have mentored hundreds of engineering students and professionals. I can say without hesitation that Olusegun's engineering capability and systems-level thinking place him in the top tier of talent I have encountered. He does not simply use technologies; he designs and builds complex systems from first principles that solve problems no one else has solved yet—and he has done so across machine learning, blockchain, cryptography, cross-platform application development, and mathematical research, entirely without institutional support.

I recommend him for the Global Talent Visa without reservation.

Yours faithfully,

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